

IA Mathematics for Natural Sciences B:

Examples Sheet 3 - Complex numbers and hyperbolic functions

Please send all comments and corrections to jmm232@cam.ac.uk.

Questions marked with a (*) are difficult and should not be attempted at the expense of the other questions.

Complex number basics

1. By expanding in real and imaginary parts, show that for any complex numbers z, w , we have:

$$\operatorname{Re}(z) = \frac{z + z^*}{2}, \quad \operatorname{Im}(z) = \frac{z - z^*}{2i}, \quad (z + w)^* = z^* + w^*, \quad (zw)^* = z^*w^*, \quad \left(\frac{z}{w}\right)^* = \frac{z^*}{w^*}.$$

Verify these results when $z = 1 + i$ and $w = 2 - 5i$.

2. (a) Factorise, and hence solve, the following polynomial equations:

$$(i) z^2 + 1 = 0; \quad (ii) z^2 + (1 - i)z - i = 0; \quad (iii) z^3 - (4 + 2i)z^2 + (4 + 5i)z - (1 + 3i) = 0.$$

- (b) Prove that the solutions to a polynomial equation with real coefficients come in complex conjugate pairs. Deduce that if such an equation has odd degree, it must have at least one real root.
(c) How does the result of part (b) relate to the equations in part (a)?

Modulus and argument of a complex number

3. (a) Define the *modulus* and (*principal*) *argument* of the complex number z . Find the modulus and (*principal*) argument of the complex numbers $z = 2 + i$ and $w = 3 + 4i$.
(b) By writing z, w in modulus-argument form, show that:

$$|zw| = |z||w|, \quad \arg(zw) = \arg(z) + \arg(w), \quad \left|\frac{z}{w}\right| = \frac{|z|}{|w|}, \quad \arg\left(\frac{z}{w}\right) = \arg(z) - \arg(w).$$

Hence give geometrical interpretations of complex number multiplication and division. Verify these results for the complex numbers z, w from part (a).

4. By considering the product of the complex numbers $1 + iA, 1 + iB$, derive a formula expressing the sum $\arctan(A) + \arctan(B)$ as a single arctangent.
5. Let $Q = zw^*$, where z, w are complex numbers.
(a) Give a geometrical interpretation of the real and imaginary parts of Q , if z, w are regarded as two-dimensional vectors. By considering a vector area, deduce that the area of the triangle formed by the points $0, z, w$ in the complex plane is $\frac{1}{2} |\operatorname{Im}(zw^*)|$.
(b) Show that Q is invariant if z, w are both rotated through the same angle, and relate your result to your previous geometrical results.

Sketching complex loci

6. By setting $z = x + iy$, identify and sketch the loci: (a) $\operatorname{Re}(z^2) = \operatorname{Im}(z^2)$; (b) $\operatorname{Im}(z^2) = -iz^2$.
7. (*Circles, lines, and half-lines*) Without setting $z = x + iy$, sketch the loci:

$$(a) |z - 2 - i| = 6; \quad (b) |z^* - i| = 1; \quad (c) |z - 2| = |z + i|; \quad (d) \arg(z) = \pi/2; \quad (e) \arg(z^*) = \pi/4.$$

If z lies on the locus in part (a), what are the loci of the complex variables $u = z + 5 - 8i$, $v = iz + 2$, and $w = \frac{3}{2}z + \frac{1}{2}z^*$?

8. (a) Show that $|z + k|^2 = zz^* + kz^* + k^*z + kk^*$.

Now consider the locus $azz^* + bz^* + b^*z + c = 0$, where $a, c \in \mathbb{R}$ and $b \in \mathbb{C}$ are constants.

- (b) Without setting $z = x + iy$, use Question 1 to show that the locus is a line when $a = 0$.
(c) Without setting $z = x + iy$, use (a) to show that the locus is a circle, point, or empty, when $a \neq 0$.
(d) Hence, without setting $z = x + iy$, sketch the loci: (i) $z + z^* - 3 = 0$; (ii) $|z - 2| = 3|z - 1|$.
(e) (*) Generalising part (d)(ii), prove that the locus $|z - \alpha| = \lambda|z - \beta|$ is always a circle when $\lambda \neq 1$, and find its centre and radius. [This form of a circle is called the Apollonian form of a circle.]
9. (Two conic sections) Sketch the loci: (a) $|z - 4| + |z^* - i| = 3$; (b) $|z - 4| - |z^* - i| = 3$.

Exponentials, logarithms, and powers

10. State Euler's formula for the complex exponential $e^{i\theta}$. Hence, express each of the following complex exponentials in Cartesian form:
11. Find the real and imaginary parts, and the modulus and argument, of the expression $e^{i\omega t} / (R + iX)$, where ω, t, R and X are real. If you're taking Part IA Physics, what might these constants represent?
12. The complex logarithm, $\log(z)$, is defined to satisfy $e^{\log(z)} = z$ for all z .
(a) By setting $\log(z) = u + iv$, where u, v are real, find u and v in terms of $|z|$ and $\arg(z)$. Hence show that there are infinitely many possible values of the complex logarithm.
(b) Using the complex logarithm, find all complex numbers z satisfying: (i) $e^{2z} = -1$; (ii) $e^{z^*} = i + 1$.
13. The complex power z^w is defined by $z^w := e^{w \log(z)}$.
(a) By considering the cases when w is an integer, rational, irrational, or complex number separately, explain when complex powers are multi-valued.
(b) Find all possible values of: (i) $i^{1/3}$; (ii) i^i ; (iii) $i^{(i^i)}$; (iv) $(i^i)^i$.

Trigonometry with complex numbers

14. State and prove De Moivre's theorem. Use it to solve the equation $16 \sin^5(\theta) = \sin(5\theta)$.
15. Using Euler's formula, show that $\sin(\theta) = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$, and $\cos(\theta) = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$. Hence express $\sin^5(\theta)$ in terms of $\sin(\theta), \sin(3\theta), \sin(5\theta)$.
16. Find all solutions of the equation $\cos(z) = 2$ in terms of logarithms.
17. Let $\theta \neq 2p\pi$ for $p \in \mathbb{Z}$. Show that $\sum_{n=0}^{N-1} \cos(n\theta) = \frac{\cos((N-1)\theta/2) \sin(N\theta/2)}{\sin(\theta/2)}$. What if $\theta = 2p\pi$?

Hyperbolic functions

18. State the definitions of $\sinh(x)$, $\cosh(x)$, and hence show that $\sinh(ix) = i \sin(x)$, $\cosh(ix) = \cos(x)$. Using this property, explain how a trigonometric identity may be transformed into a hyperbolic identity. Thus, give the hyperbolic analogues of the formulae:
- $$\cos^2(x) + \sin^2(x) = 1, \quad \cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y), \quad \tan(x+y) = \frac{\tan(x) + \tan(y)}{1 - \tan(x)\tan(y)}.$$
19. Express $\sinh^{-1}(x)$, $\cosh^{-1}(x)$ and $\tanh^{-1}(x)$ as logarithms, justifying any sign choices you make.
20. Find all solutions to the equation $\cosh(z) = i$.